Digital Logic final test solution

1.

ABCD	$Y_1'Y_0' = S_1S_0$	$Z_3Z_2Z_1Z_0$	M by CD	F
0000	11	1000	Z_0	0
0001	11	1000	Z_1	0
0010	10	0100	Z_2	1
0011	10	0100	Z_3	0
0100	01	0010	Z_0	0
0101	01	0010	Z_1	1
0110	01	0010	Z_2	0
0111	01	0010	Z_3	0
1000	00	0001	Z_0	1
1001	00	0001	Z_1	0
1010	00	0001	Z_2	0
1011	00	0001	Z_3	0
1100	00	0001	Z_0	1
1101	00	0001	Z_1	0
1110	00	0001	Z_2	0
1111	00	0001	Z_3	0

By Karnaugh map

F = A'B'CD' + A'BC'D + AC'D'

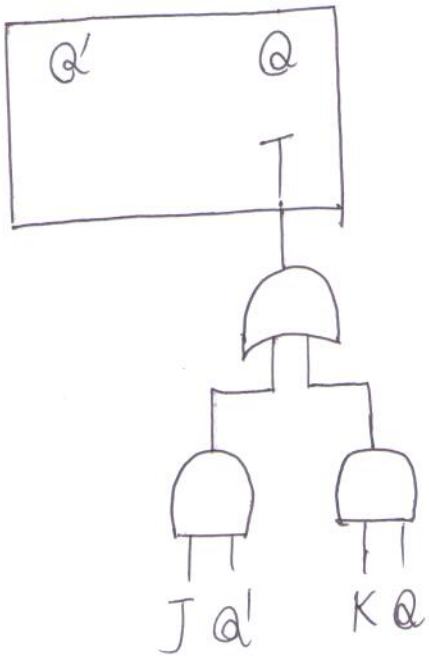
AB \ CD	00	01	11	10
00	0	0	0	1
01	0	1	0	0
11	1	0	0	0
10	1	0	0	0

$\boxed{2}$ JK ← T → design inputs of T-Flip.

JKQ	Q^T	T
000	0	0
001	1	0
010	0	0
011	0	1
100	1	1
101	1	0
110	1	1
111	0	1

JK \ Q	T = 0	T = 1
00	0	0
01	0	1
11	1	1
10	1	0

$\Rightarrow J = JQ' + KQ.$



$\boxed{3}$

Find D_A, J_B. K_B for all XAB, (000 ~ 111).

and then find A^+B^+.

	AB	A^+B^+		Z	
		X=0	X=1	X=0	X=1
S_0	00	01	00	0	0
S_1	01	00	11	1	0
S_2	11	10	11	1	1
S_3	10	00	01	0	1

4.

Set $S_0 = 00$, $S_1 = 01$, $S_2 = 11$ $S_3 = 01$. (AB).

S_i	$AB X_1 X_2$	$A^+ B^+$	$J_A K_A$	$J_B K_B$
S_0	0000	10	1 X	0 X
	0001	11	1 X	1 X
	0010	01	0 X	1 X
	0011	00	0 X	0 X
S_1	0100	00	0 X	X 1
	0101	01	0 X	X 0
	0110	11	1 X	X 0
	0111	10	1 X	X 1
S_3	1000	11	X 0	1 X
	1001	11	X 0	1 X
	1010	01	X 1	1 X
	1011	00	X 1	0 X
S_2	1100	10	X 0	X 1
	1101	00	X 1	X 1
	1110	01	X 1	X 0
	1111	01	X 1	X 0

$Q \to Q^+$	J K
$0 \to 0$	0 X
$0 \to 1$	1 X
$1 \to 0$	X 1
$1 \to 1$	X 0

J_A

AB \ X_1X_2	00	01	11	10
S_0 00	1	1	0	0
S_1 01	0	0	1	1
S_2 11	X	X	X	X
S_3 10	X	X	X	X

K_A

AB \ X_1X_2	00	01	11	10
00	X	X	X	X
01	X	X	X	X
11	0	1	1	1
10	0	0	1	1

AB \ X_1X_2	00	01	11	10
00	0	1	0	1
01	X	X	X	X
11	1	1	0	1
10	X	X	X	X

1-1-4 가지.

AB \ X_1X_2	00	01	11	10
00	X	X	X	X
01	1	0	1	0
11	X	X	X	X
10	1	1	0	0

1-2-2

$\boxed{4}$

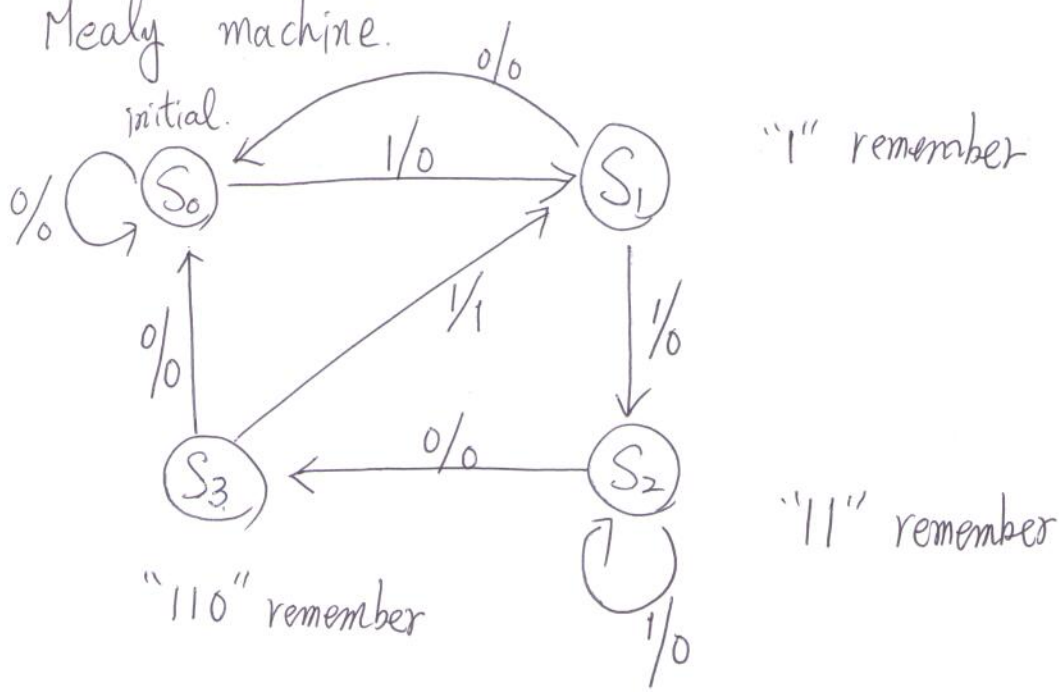
$$\begin{cases} J_A = B'X_1' + BX_1 \\ K_A = X_1 + BX_2 \end{cases}$$

$$J_B = X_1'X_2 + X_1X_2' + \begin{cases} BX_1' \\ AX_1' \\ BX_2' \\ AX_2'. \end{cases}$$

$$K_B = X_1'X_2' + \begin{cases} B'X_1' \\ AX_1' \end{cases} + \begin{cases} A'X_1X_2 \\ BX_1X_2. \end{cases}$$

5. "1101".

(1) Mealy machine.



(2) $S_0 = 00$, $S_1 = 01$, $S_2 = 11$, $S_3 = 10$.

XAB	A^+B^+	Z	J_A K_A	J_B K_B
000	00	0	0 X	0 X
001	00	0	0 X	X 1
010	00	0	X 1	0 X
011	10	0	X 0	X 1
100	01	0	0 X	1 X
101	11	0	1 X	X 0
110	01	1	X 1	1 X
111	11	0	X 0	X 0

JK Fn.

$0 \to 0$: 0X
$0 \to 1$: 1X
$1 \to 0$: X1
$1 \to 1$: X0

J_A

AB \ X	0	1
00	0	0
01	0	1
11	X	X
10	X	X

$J_A = XB$

K_A

AB \ X	0	1
00	X	X
01	X	X
11	0	0
10	1	1

$K_A = B'$

J_B

AB \ X	0	1
00	0	1
01	X	X
11	X	X
10	0	1

$J_B = X$

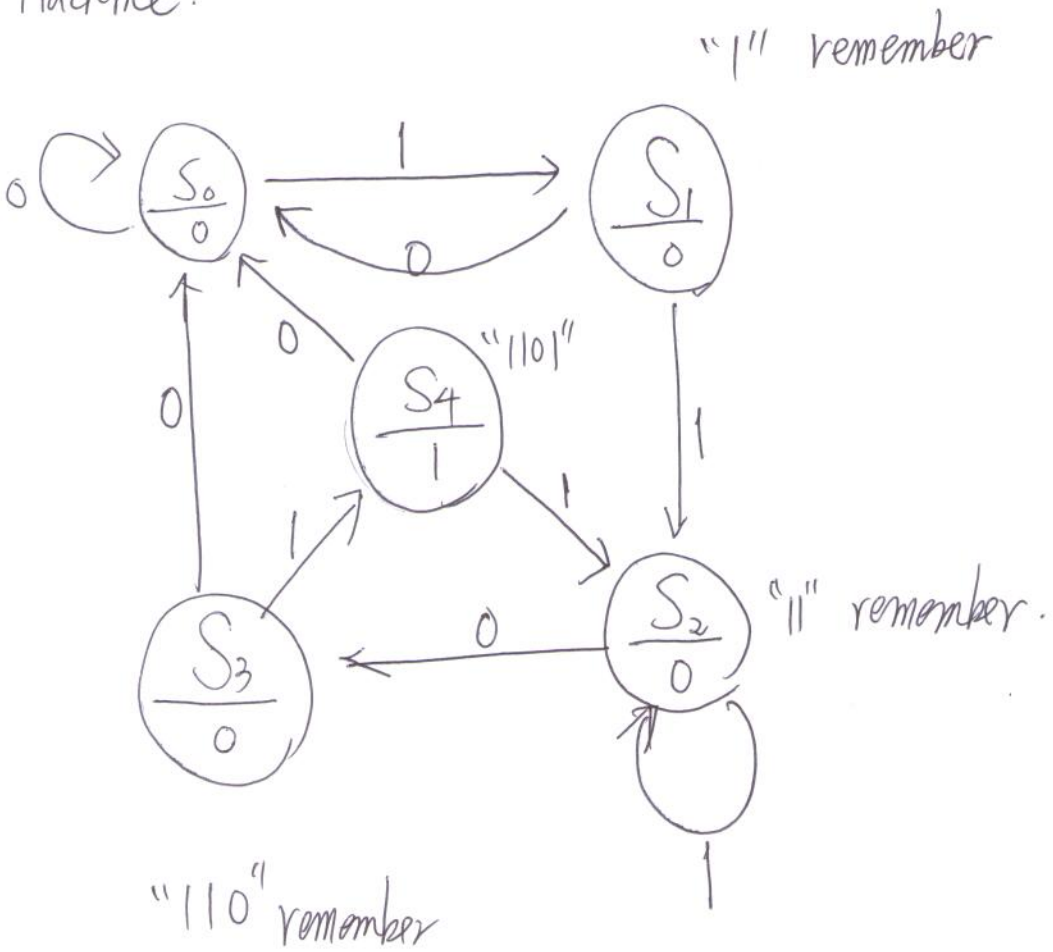
K_B

AB \ X	0	1
00	X	X
01	1	0
11	1	0
10	X	X

$K_B = X'$

$Z = XAB'$

5

No. 7

(3) Moore Machine.